ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

CIVIL TECHNOLOGY

THEORY PAPER

MARKS: 100 **TIME: 2 hours**

Topics: safety, materials, tools and equipment, construction and quantities, and graphics. Suitable for Civil Services, Construction and Woodworking learners.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions. Show ALL calculations with units. Make neat sketches where asked.

QUESTION 1: MULTIPLE CHOICE

1.1 The law that deals with health and safety in South African workplaces is the …

A Labour Relations Act. B Occupational Health and Safety Act.
C Skills Development Act. D Basic Conditions of Employment Act.

1.2 Concrete is made of cement, water, sand and …

A lime. B stone (coarse aggregate).
C clay. D timber.

1.3 The tool used to check that a wall is vertical is a …

A spirit level or plumb bob. B try square.
C trowel. D chisel.

1.4 On a drawing with a scale of 1 : 50, 1 mm on the drawing represents …

A 5 mm. B 50 mm.
C 500 mm. D 5 m.

1.5 Which of the following is a hardwood?

A Pine B Kiaat
C Fibreboard D Chipboard

(5 × 2)
[10]

QUESTION 2: SAFETY

2.1 Name THREE duties of an employer under the Occupational Health and Safety Act. (3)

2.2 Name the protective equipment needed when: (a) using an angle grinder; (b) mixing cement; (c) working on a roof. (3)

2.3 What does each colour mean on a safety sign: red, blue, yellow and green? (4)

2.4 Describe the first aid for a worker with a deep cut on the hand. (3)

2.5 Why must a workshop be well ventilated? (2)

[15]

QUESTION 3: MATERIALS

3.1 A concrete mix of 1 : 2 : 3 (cement : sand : stone) is used to make 0,6 m³ of dry mix. Calculate the volume of each material. (4)

3.2 What is meant by *curing* concrete, and why is it important? (3)

3.3 Give TWO advantages of timber as a building material. (2)

3.4 Name THREE defects found in timber. (3)

3.5 Name TWO types of bricks. (2)

3.6 What is the function of a damp-proof course (DPC) in a wall? (2)

3.7 Give TWO methods of protecting (preserving) timber. (2)

3.8 Why is steel reinforcement placed in concrete? (2)

[20]

QUESTION 4: TOOLS AND EQUIPMENT

4.1 State the use of each tool: claw hammer, tenon saw, brick trowel, cold chisel, builder's line. (5)

4.2 Give THREE rules for looking after electrical power tools. (3)

4.3 Explain how to check whether a spirit level is accurate. (3)

4.4 Name TWO wood joints and give ONE use of each. (4)

[15]

QUESTION 5: CONSTRUCTION AND QUANTITIES

5.1 What is the purpose of a foundation? (2)

5.2 Calculate the volume of concrete needed for a strip foundation 12 m long, 600 mm wide and 200 mm deep. (3)

5.3 A half-brick wall is 6 m long and 2,4 m high. 55 bricks are needed per m².

5.3.1 Calculate the number of bricks needed. (4)

5.3.2 Add 5% for breakages. How many bricks must be ordered? (2)

5.3.3 Bricks cost R4,50 each. Calculate the cost of the bricks ordered. (2)

5.4 Name FOUR parts of a pitched timber roof. (4)

5.5 What is the purpose of a lintel? (2)

5.6 Give THREE reasons for compacting the soil before building. (3)

5.7 Explain the 3-4-5 method for setting out a right angle on a building site. (3)

[25]

QUESTION 6: GRAPHICS

6.1 What is each type of line used for on a building drawing: (a) a continuous thick line; (b) a dashed (broken) line; (c) a chain line (long dash, short dash)? (6)

6.2 Write out these abbreviations in full: DPC, FFL, NGL. (3)

6.3 A wall is 4 m long. How long must it be drawn at a scale of 1 : 50? Give your answer in mm. (2)

6.4 Name the THREE main views used in orthographic projection. (3)

6.5 Why are scales used on building drawings? (1)

[15]

TOTAL: 100

END OF QUESTION PAPER